RASCL
WP 3: Robot Control

Prof. Dr.-Ing. Mike Barth

Prof. Dr.-Ing. Sören Hohmann

Eric Wagemann, Linus Witucki, Pirmin Breisacher

Why Minimum-Jerk in Motion-Planning?

- Tracking control performance is inversely influenced by Jerk [1,2]
- Reduction of induced vibrations [1,2]
- Reduced component wear [1,2]
- Reduced energy consumption [2]
- Increased task success rate [2]
- Human-robot collaboration demands MJ-trajectories [3] ...
  - ... for safety reasons
  - ... to reduce cognitive load on the human operator

[1] Oliveira et al., A General Framework for Optimal Tuning of PID-like Controllers for Minimum Jerk Robotic Trajectories, Journal of Intelligent & Robotic Systems, 2020

[2] Lee et al., On the Performance of Jerk-Constrained Time-Optimal Trajectory Planning for Industrial Manipulators, The IEEE International Conference on Robotics and Automation, 2024, https://www.youtube.com/watch?v=9p0bmgBnL2Q

[3] Lozer et a., Planning optimal minimum-jerk trajectories for redundant robots, Robotics and Autonomous Systems, 2025

Task Description - Summary

- Control the manipulator arm using two approaches:
  - Task 1: *Offline* motion planning
  - Task 2: *Online* motion planning
- Solve Pick-and-Place tasks with 3D-printed cubes
  - **Restrictions on cube placement apply**
- Submission guidelines: see the task description document
  - **Adhere to the naming conventions**
- **Before working on the actual hardware, validate your approaches in simulation**

Task description – AI use

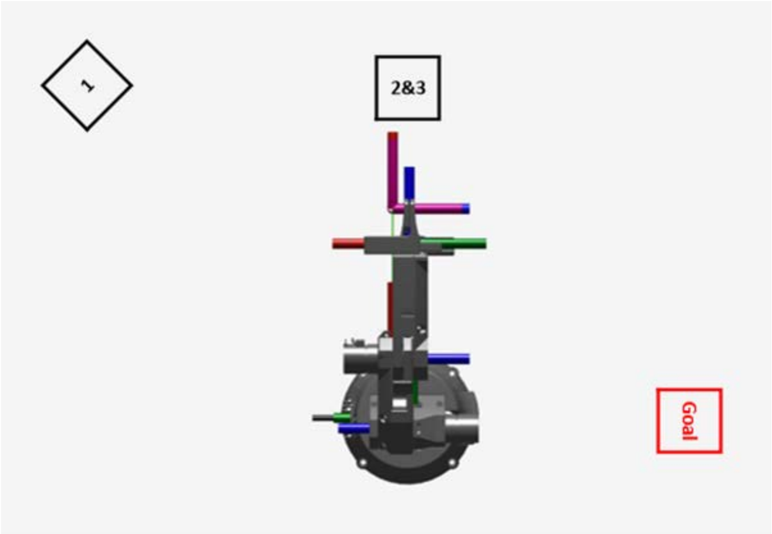
- Using generative AI (GAI) is encouraged throughout the whole work package
- Why?
  - There is a shift in the required skills:
    - Artifact creation itself can increasingly be automated (vibe coding for example, see further readings below)
    - Thorough specification, validation and verification of the output become increasingly important (see further readings below)
  - Therefore, we want you to achieve proficiency in the use of GAI in engineering.
    - Reflect your approach in prompting, **critically assess the models output**, and adjust your prompts accordingly: adjustments in the prompts contents but also the approach of asking – suggestive questions vs. open questions
    - Bring your own ideas into the interaction with the model, gain the ability to assess how much foundational knowledge must be acquired before using AI
    - How far can you trust created artifacts and answers, what needs thorough investigation and maybe even additional literature research?
- Further readings:
  - https://www.iese.fraunhofer.de/blog/generative-ai-in-se-and-ai-orchestrated-sdlc-retrospective/
  - https://blogs.sw.siemens.com/art-of-the-possible/generative-engineering-and-the-role-of-simulation/#section_3
  - https://dev.to/ziv_kfir_aa0a372cec2e1e4b/why-the-v-model-is-the-natural-way-to-work-with-ai-coding-agents-17g6
  - https://www.linkedin.com/pulse/die-erste-pneutronic-maschine-l%C3%A4uft-mit-100-code-marcel-gloor-lgcee?utm_source=share&utm_medium=member_android&utm_campaign=share_via

Task description – Task 1

- Pre-compute minimum-jerk trajectories to move the cubes. Every robot movement has to be based on such a trajectory.
- Your node shall accept these offline-trajectories and execute the movement.
- Use the MCs CSP mode and feed it with a position trajectory. Do not implement an effort interface in your top-level control.
- There are no further restrictions on how you implement your solutions.
- Find details in the task description document.

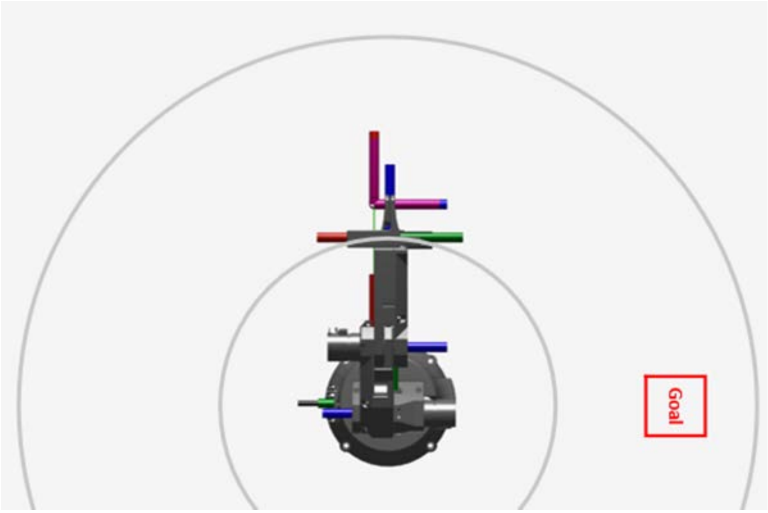
Task Description – Task 1: Cube Placement

- Two cubes must be stacked but they must be moved individually.
- Each cube must be moved to a new place.
- Every stack must be placed on a different radius.
- Cube placement must be as shown in the picture:
  - Cube 1 must be placed at left of the robot’s reference switch.
  - The cubes 2 and 3 are near the reference switch. Cubes 2 is at the bottom and cube 3 is on top.
  - The goal position is on the right side.
- All three cube positions must differ significantly regarding the radius
  - $r_1 > r_{2/3} > r_{goal}$
- The cubes must be stacked in order 1-2-3 with cube 1 at the bottom, cube 2 in the middle, and cube 3 on top.



Task description – Task 2

- Pick and place an arbitrary number of cubes. The positions of the cubes are not known in advance but will be published to a topic at runtime
- For task validation: contact the lab tutors and fix a date for supervised validation. For details see the task description document.
- Placement goal: as in task 1
- No obstacles
- Find details in the task description document



Organizational Topics

- Consulting hours
  - Question submission deadline: 24h prior
  - No questions = no consulting hour
  - Dates:
    - Tuesday, July 7, 10.30h – 11.30h
    - Tuesday, July 21, 10.30h – 11.30h

- Evaluation
  - Link to the evaluation will be published in a few weeks on ILIAS
  - Please evaluate as this helps us improve

- Exam
  - Date: August 12, 09:00h at building 20.21 Pool E and Pool H
  - Recall: no problem **solving**, we want to assess your understanding and your reflection abilities

IRS